

Dolly POS — Master Operations & Troubleshooting Guide

Complete Step-by-Step Store Counter Handbook, Hardware Setup & Recovery Manual

Store: Dolly Toys & Kids Wear	Software: Dolly POS Enterprise Edition
Location: Agra Road, Near MG Statue, Dhule	Release Version: Production v1.0.4
Operating Since: 17 February 2002	Document Date: September 2026
Primary Thermal Printer: 80mm ESC/POS Roll	Secondary Barcode Printer: XP-D451B / 2-Up Labels

Chapter	Topic & Coverage
1. Daily Routine	How to start POS, terminal commands, login & system architecture
2. Password Recovery	Admin vs Staff accounts, emergency password reset script & permissions
3. Billing & Checkout	Barcode scanning, 1-click Speed Dials, unlisted items, discounts, UPI QR, Parked bills
4. Thermal Printer Setup	80mm/58mm driver installation, Chrome Print settings (0 Margins), fixing light/fade print
5. Returns & Exchanges	Return search by bill number, replacement items, net calculations, defect handling
6. Inventory & Barcodes	Adding garments/toys, size/color matrices, generating barcodes & printing 2-Up stickers
7. Khata & Vendor Ledgers	Customer credit ledger, 1-click WhatsApp reminders, vendor purchases & settlements
8. Dead Stock Advisor	1-Year stagnant stock liquidation, 17th Feb anniversary automation & smart discounts
9. Backup & Disaster Recovery	Where dolly_pos.db is stored, 1-click Google Drive/Pendrive backup, restoring in 5 mins
10. WhatsApp Bill PDF	Direct wa.me 1-click bill links vs Official WhatsApp Cloud API integration roadmap
11. Troubleshooting Master	Port 8000/5173 occupied, DB locked, printer spooler error, keyboard shortcuts cheat sheet

Chapter 1: System Overview & Daily Startup Routine

1.1 System Architecture

Dolly POS is an enterprise-grade, local-first retail software built for 100% offline reliability. It runs with two independent high-performance components:

- **Backend API Engine (FastAPI + SQLite):** Runs on port 8000. Manages the relational database, inventory transactions, bill sequence generation, and financial audit records.
- **Frontend Counter UI (React + TypeScript + Vite):** Runs on port 5173. Renders the cashier checkout screen, speed dials, thermal receipt builder, and live business analytics.

1.2 How to Start the Application Every Morning

Open Windows PowerShell or Command Prompt in the project folder and start the services:

```
Step 1: Start Backend API
cd "c:\Users\SomeshBang\Desktop\Antigravity DollyPos\backend"
.\venv\Scripts\activate
uvicorn app.main:app --host 0.0.0.0 --port 8000 --reload
```

```
Step 2: Start Frontend POS (in a second terminal)
cd "c:\Users\SomeshBang\Desktop\Antigravity DollyPos\frontend"
npm run dev
```

Step 3: Open in Chrome Browser

Open your browser and navigate to: **http://localhost:5173**

1.3 1-Click Desktop Launcher (.BAT)

To start both servers with a single double-click from the desktop without typing commands, use start_pos.bat in the root directory:

```
@echo off
start cmd /k "cd backend && .\venv\Scripts\activate && uvicorn app.main:app --port 8000"
start cmd /k "cd frontend && npm run dev"
timeout /t 3
start http://localhost:5173
```

Why localhost stays logged in: The app securely stores active login tokens in the browser's localStorage. In the packaged desktop executable version, closing the application window can be set to require password/PIN entry on relaunch.

Chapter 2: User Access, Permissions & Password Recovery

2.1 User Roles & Access Control

- **ADMIN / OWNER:** Full store access — can edit product purchase & selling prices, view profit/loss reports, manage vendor ledgers, change GST settings, reset staff passwords, and delete defective stock.
- **CASHIER / STAFF:** Restricted counter access — can create bills, scan barcodes, process returns/exchanges, collect cash/UPI, and park bills. Critical financial reports and vendor cost prices remain hidden.

2.2 Emergency Password Reset (If Locked Out)

If the admin password is forgotten, run this emergency Python reset command in your backend directory:

```
cd backend
python -c "from app.core.database import SessionLocal; from app.models.user import User; from app.core.security import get_password_hash; db = SessionLocal(); u = db.query(User).filter(User.username == 'admin').first(); u.hashed_password = get_password_hash('admin123'); db.commit(); print('Admin password successfully reset to: admin123')"
```

After running the command, log in with username: admin and password: admin123, then change your password inside the Settings & Staff tab.

Chapter 3: Master Billing & Checkout Workflow

3.1 Item Entry Methods (3 Fast Options)

- Barcode Scanner (Instant):** Point laser scanner at tag. Item, size, rate, and tax are added to cart with audio confirmation chime.
- 1-Click Speed Dials (Top Sellers):** Press **F2** or click 'Speed Dials' to instantly pick un-barcoded toys, socks, hankies, hairclips, and daily accessories.
- Unlisted Item Entry:** Press **F3** to sell non-catalog items on the fly by entering custom title, quantity, and rate without adding inventory.

3.2 Indian Calendar Date Sequencing (Post-Midnight Accuracy)

Bill numbers are automatically prefixed with Indian Standard Time (IST UTC+5:30): DLY-YYYYMMDD-#### (e.g. DLY-20260906-0001). Even if billing at 12:45 AM, bills increment under the correct calendar date.

3.3 Payment Modes & Dynamic UPI QR

- CASH:** Real-time change calculator (tells cashier exact amount to return to customer).
- UPI (Dynamic Instant QR):** Generates an on-screen and on-bill dynamic Bharat UPI QR code embedded with your store VPA and exact payable bill amount. Customer scans via GPay/PhonePe/Paytm without typing any number.
- SPLIT PAY:** Customer can pay part cash (e.g. ₹500) and balance via UPI (e.g. ₹750).
- CREDIT KHATA:** Records bill under customer's ledger balance with automatic credit limit warning.

3.4 Parked / Held Bills (Queue Buster)

If a customer steps aside to choose another dress, press **F6 (Hold Bill)**. The counter clears instantly for the next customer. When ready, press **F7** to retrieve the parked cart with all items intact.

Chapter 4: Thermal Receipt Printer Setup & Troubleshooting

4.1 Connecting XP-S300N / 80mm POS Receipt Printers

- Plug the USB cable into your Windows PC and power on the thermal printer.
- Download the official **80mm Series Receipt Printer Driver (POS-80)** from Xprinter / POS support.
- Select **USB Port** (e.g. USB001) and click 'Install Driver'.
- In Windows 'Printers & Scanners', set the printer as **Default Printer**.

4.2 The 1-Time Chrome Print Dialog Settings (Crucial for 0 Margins)

When the print preview pops up in Google Chrome or Edge, configure these settings once:

Chrome Print Setting	Required Value	Why it Matters
Destination	POS-80 / XP-S300N	Ensures print commands go directly to thermal roll
Paper Size	80 x 297 mm or 80 x Receipt	Matches physical 80mm roll width
Margins	"None" (or "Minimum")	Removes browser default 10mm white borders on sides & top
Headers and Footers	UNCHECK (Turn OFF)	Prevents browser from printing URL, date and page numbers
Scale	100% (Default)	Maintains crisp 1:1 font sharpness

4.3 Fixing Faded, Light or Blank Receipts

- **Paper inserted upside down:** Thermal paper is chemically coated on only one side. Scratch roll surface with your fingernail — if a black mark appears, that shiny side must face the thermal heating head.
- **Print density:** In Windows Printer Properties -> Device Settings -> Set Print Density to **Dark / High (12-14)**.
- **Cleaning thermal head:** Wipe dark thermal glass line inside printer with a cotton swab soaked in isopropyl alcohol.

Chapter 5: Return & Exchange Workflow

5.1 Step-by-Step Return / Exchange Procedure

1. Open **Returns & Exchange** tab (or press **F4**).
2. **Option A (Customer has bill)**: Enter Bill Number (e.g. DLY-20260906-0012) and click Search. All purchased items with sizes and rates appear.
3. **Option B (Walk-in Return without bill)**: Search returned item by name or scan barcode to add directly to return credit.
4. **Select Returned Items**: Tick checkbox next to product being returned. If item is damaged/torn, check '**Defective Stock**' (automatically transfers to defective quarantine instead of good stock).
5. **Add Replacement Exchange Items**: Search or scan new items customer wants to take in exchange.
6. **Automatic Settlement Math**: System calculates:
$$\text{Net Difference} = (\text{New Exchange Items Value}) - (\text{Return Credit Value})$$
 - If **Difference > 0**: System prompts cashier to **Collect Cash/UPI**.
 - If **Difference < 0**: System prompts cashier to **Refund Cash** or issue Store Credit.
 - If **Difference = 0**: Even exchange completed.
7. Click **Process Exchange** -> Prints clean 80mm **Return & Exchange Bill** voucher.

Chapter 6: Inventory & Barcode Label Sticker Printing

6.1 Adding New Garments & Toys (Continuous Fast Entry)

In **Inventory** -> **Add Product (F1)**, enter category (Boys Wear, Girls Wear, Baby Care, Toys), product title, size (e.g. 24, 26, 28, S, M, XL), color, purchase cost, and selling price (MRP). Modal supports continuous entry with automatic barcode generation.

6.2 Printing Barcode Price Tags (XP-D451B / 2-Up Rolls)

- Supports standard dual-column **50mm x 25mm** and **38mm x 25mm 2-Up** thermal transfer sticker rolls.
- Tags print with: **Shop Name, Product Name, Size/Color, Scannable Barcode, and MRP**.
- **Batch Printing**: Select multiple products or specify exact quantity (e.g. print 24 stickers for 2 dozen frocks) with 1 click.

Chapter 7: Customer Khata Ledger & Vendor Management

7.1 Customer Credit (Udhaar / Khata) Management

- Automatic auto-registration of customer phone numbers at checkout.
- Live ledger records every purchase, partial payment, and pending balance.
- **1-Click WhatsApp Payment Reminder**: Generates a polite, personalized WhatsApp message with total due balance and shop UPI QR payment link.

7.2 Vendor Purchase Ledger & Settlements

Record vendor stock invoices (Mumbai, Surat, Ahmedabad suppliers). Track invoice total, amount paid via Cheque/NEFT, and outstanding supplier liabilities.

Chapter 8: Trapped Capital (Dead Stock) & Smart Advisor

8.1 What is Trapped Capital Detection?

Garments and toys sitting in stock for over 365 days lock up cash flow. The Smart Advisor continuously aggregates the total monetary value of unsold aged stock and suggests discount markdown percentages (e.g. 20% to 40% Off) to liquidate dead stock into liquid capital.

8.2 17th February Shop Anniversary Automation

Dolly Toys & Kids Wear was founded on **17 February 2002**. Every year on February 17th, the POS automatically activates an Anniversary celebratory banner and promotional discount engine to reward loyal customers.

Chapter 9: Database Backup, Protection & Disaster Recovery

9.1 Where is Your Data Stored?

Your entire store database (all bills, products, customers, vendors, audit logs, and settings) is stored in a single secure local file:

```
c:\Users\SomeshBang\Desktop\Antigravity DollyPos\backend\dolly_pos.db
```

9.2 How to Take Daily Backups (1-Click Protection)

1. Insert a USB Pen Drive or configure a Google Drive / OneDrive synced folder.
2. Copy dolly_pos.db to your backup drive at the end of each day.
3. **Automated Windows Daily Backup Script:** Create a file named backup_db.bat:

```
@echo off
set BACKUP_DIR=D:\Dolly_POS_Backups
mkdir %BACKUP_DIR% 2>nul
copy "c:\Users\SomeshBang\Desktop\Antigravity DollyPos\backend\dolly_pos.db"
"%BACKUP_DIR%\dolly_pos_%date:~10,4%%date:~4,2%%date:~7,2%.db"
echo Backup Complete!
```

9.3 How to Restore Data on a Fresh Computer (5-Minute Recovery)

If your counter PC crashes or hard drive is replaced:

1. Copy project folder Antigravity DollyPos onto the new computer.
2. Paste your latest backup file into backend/ and rename it to dolly_pos.db.
3. Launch POS — all historical bills, customers, and stock are 100% restored immediately.

Chapter 10: WhatsApp PDF Billing Integration Roadmap

10.1 Current 1-Click WhatsApp Sharing

On checkout receipt screen, clicking '**Send Bill on WhatsApp**' automatically formats a crisp itemized invoice with store address, bill number, GST breakdown, item details, and total, opening WhatsApp Web or WhatsApp Desktop with customer phone pre-filled.

10.2 Steps for Official WhatsApp Cloud API (Automated Background PDF Delivery)

To have POS automatically send a PDF directly to customer phones without opening WhatsApp manually:

1. Register a **Meta for Developers** business account at developers.facebook.com.
2. Activate **WhatsApp Business Cloud API** and link dedicated store phone number.
3. Obtain permanent **Bearer Token** and **Phone Number ID**.
4. Insert token into backend/app/services/whatsapp_service.py to trigger automated PDF dispatch immediately upon checkout completion.

Chapter 11: Comprehensive Troubleshooting & Emergency FAQ

Problem / Error Message	Root Cause	Immediate Solution
Port 8000 / 5173 Already in Use	Previous server process did not terminate cleanly when terminal closed.	Run in PowerShell: Stop-Process -Name python, node -Force Then restart servers.
Network Error / Failed to Fetch	Backend FastAPI server is stopped or crashed.	Check backend terminal window. Ensure uvicorn app.main:app is running and shows 'Application startup complete'.
Database is Locked (sqlite3.OperationalError)	Another application (e.g. DB Browser) has opened database with exclusive write lock.	Close DB Browser or restart backend service. Dolly POS uses SQLite WAL mode for high concurrency.
Printer Prints Extra White Margin on Left / Right	Chrome Print Margins set to 'Default' instead of 'None'.	Press Enter to print -> In Chrome Print window click 'More settings' -> Set Margins: None and uncheck 'Headers and footers'.
Bill Number sequence mismatch	Timezone lag across midnight.	Resolved: System uses Indian Standard Time (IST UTC+5:30) generation natively.
Modal / Popup won't close	Mouse focus lost.	Press Escape (Esc) key. Every dialog and modal in Dolly POS dismisses instantly with Esc.

11.2 POS Keyboard Shortcuts Master Reference Sheet

Key	Action	Key	Action
F1	Add Product / Inventory Entry	F6	Hold / Park Current Bill
F2	Open Speed Dials Grid	F7	Recall Parked / Held Bills
F3	Quick Unlisted Item Entry	F9	Procurement Lost Demand Log
F4	Returns & Exchange Screen	Enter	Quick Pay / Print Thermal Bill
F5	Barcode Label Generator	Esc	Close Active Modal / Cancel

End of Operational Manual • Dolly Toys & Kids Wear (Since 2002)